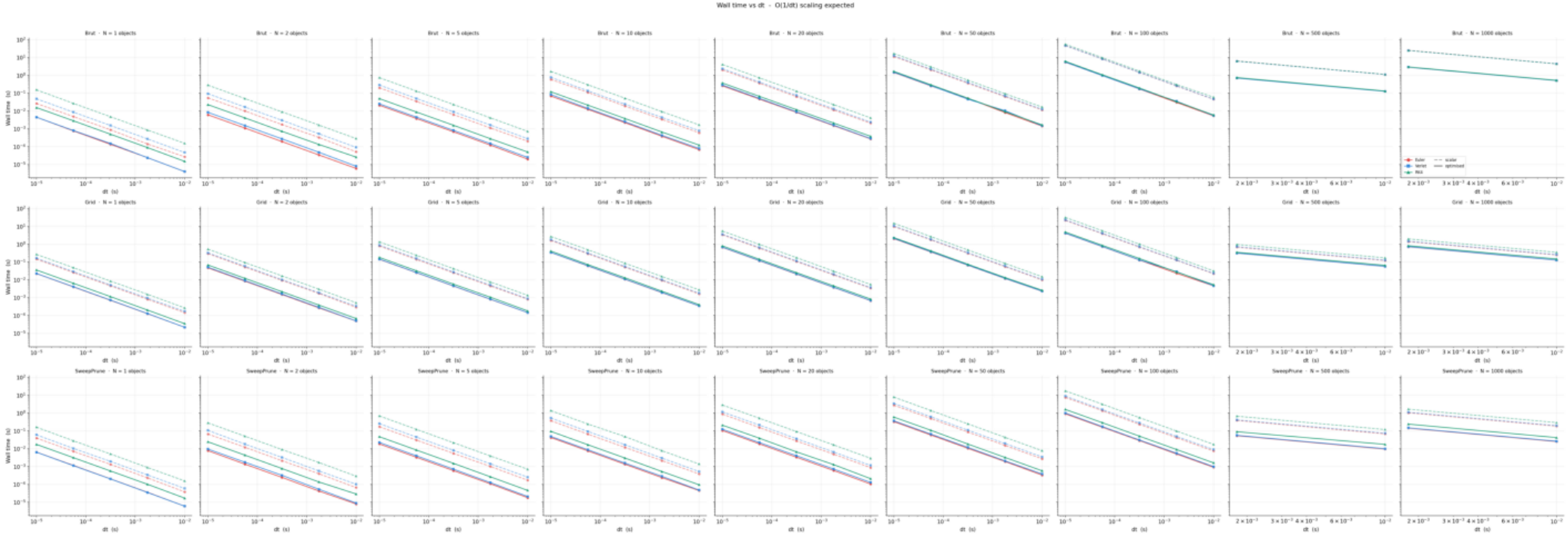
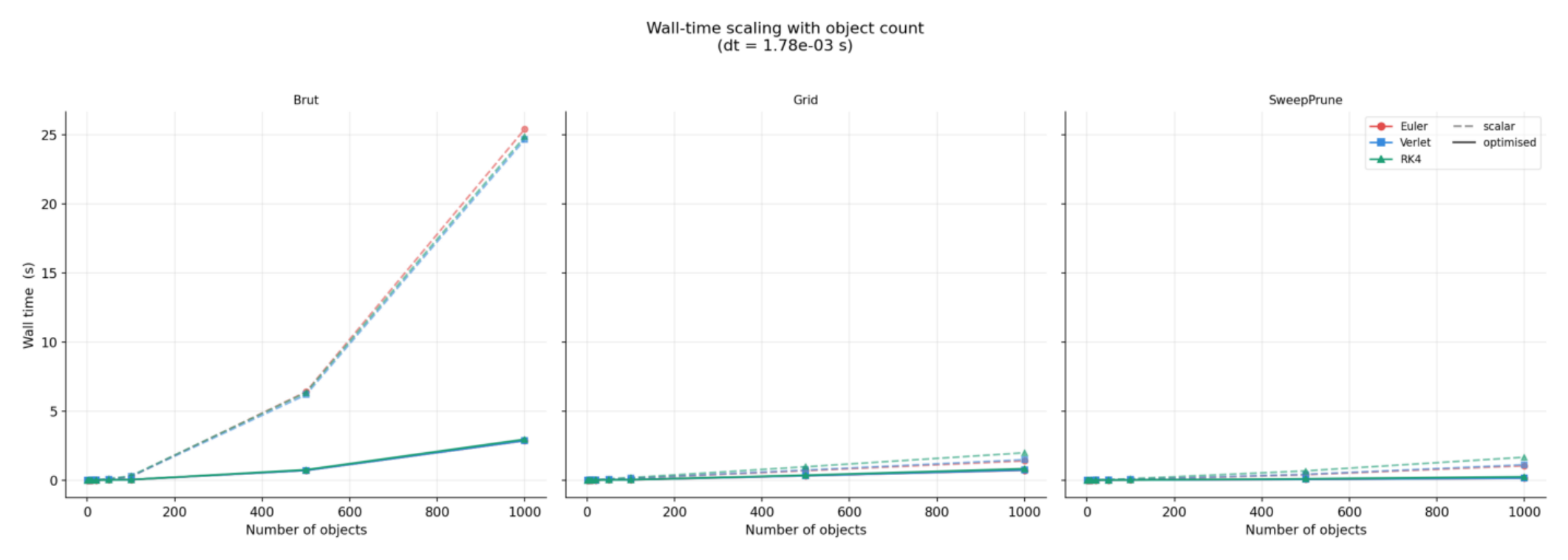
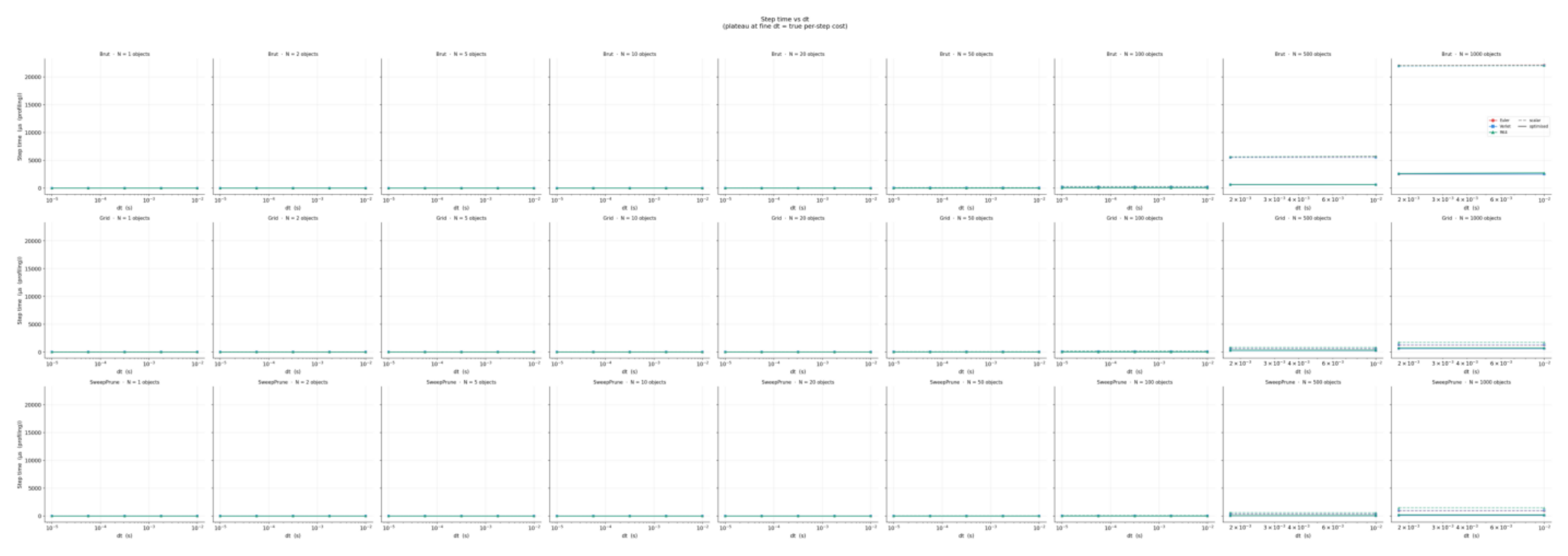
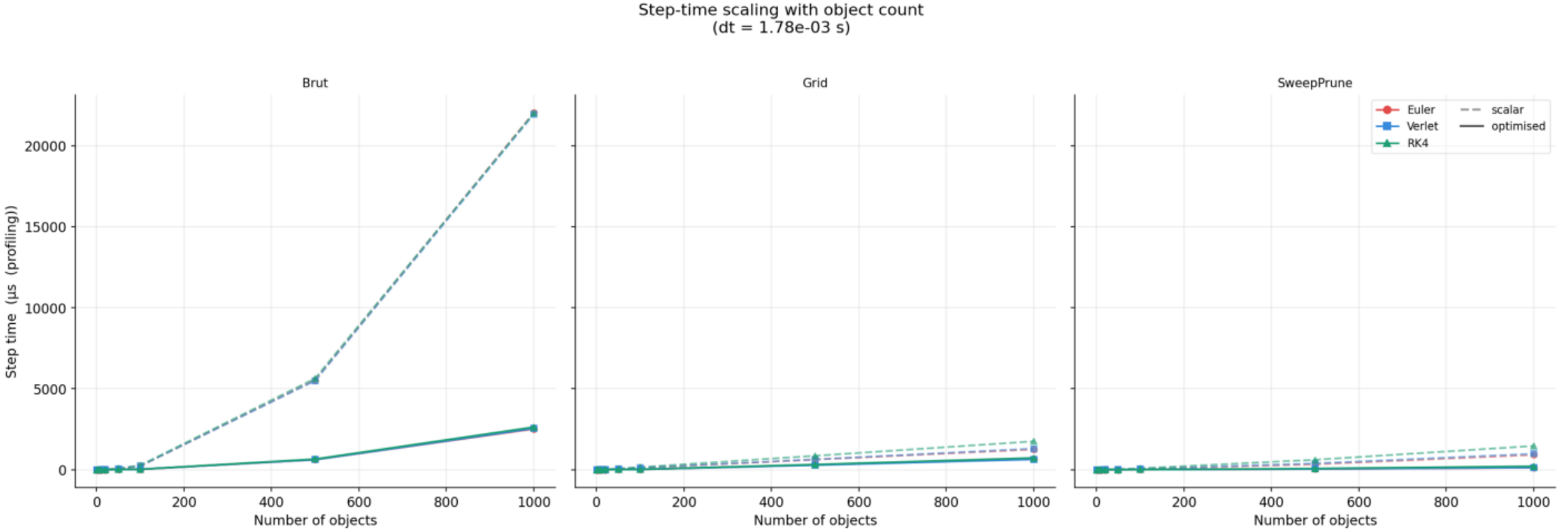
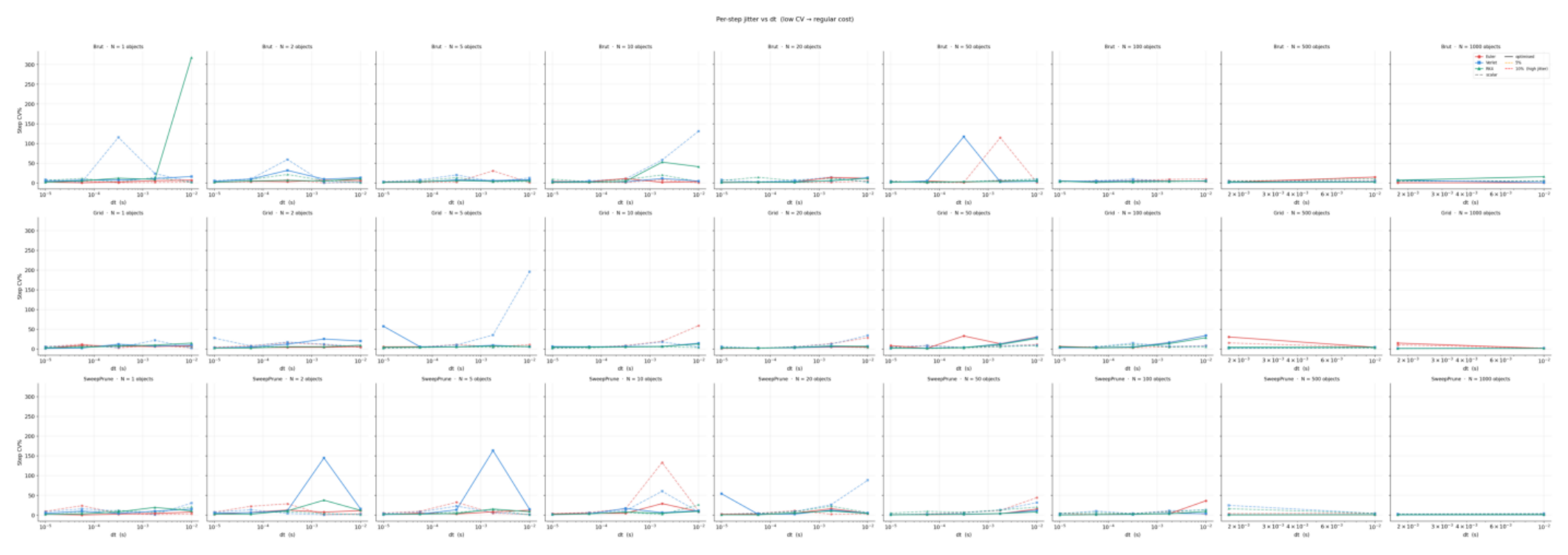
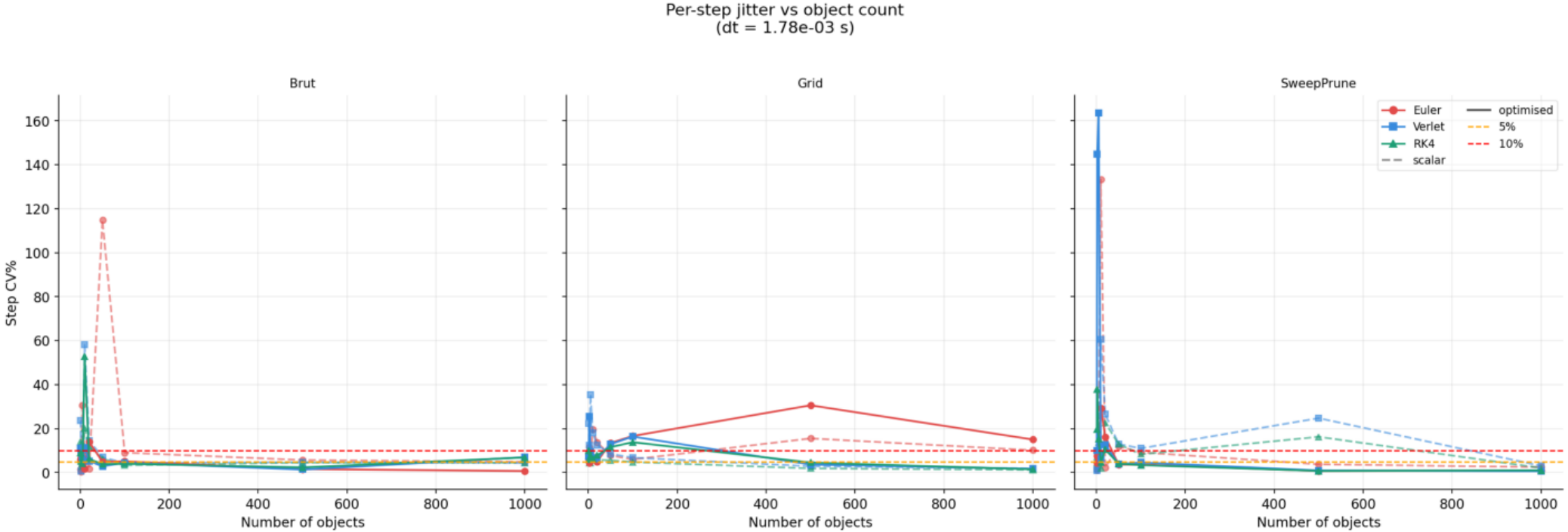

Wall time vs dt – O(1/dt) scaling expected
Brut · N = 1 objects
Brut · N = 2 objects
Brut · N = 5 objects
Brut · N = 10 objects
Brut · N = 20 objects
Brut · N = 50 objects
Brut · N = 100 objects
Brut · N = 500 objects
Brut · N = 1000 objects
Grid · N = 1 objects
Grid · N = 2 objects
Grid · N = 5 objects
Grid · N = 10 objects
Grid · N = 20 objects
Grid · N = 50 objects
Grid · N = 100 objects
Grid · N = 500 objects
Grid · N = 1000 objects
SweepPrune · N = 1 objects
SweepPrune · N = 2 objects
SweepPrune · N = 5 objects
SweepPrune · N = 10 objects
SweepPrune · N = 20 objects
SweepPrune · N = 50 objects
SweepPrune · N = 100 objects
SweepPrune · N = 500 objects
SweepPrune · N = 1000 objects
Wall time (s)
dt (s)
Euler
Verlet
RK4
scalar
optimized

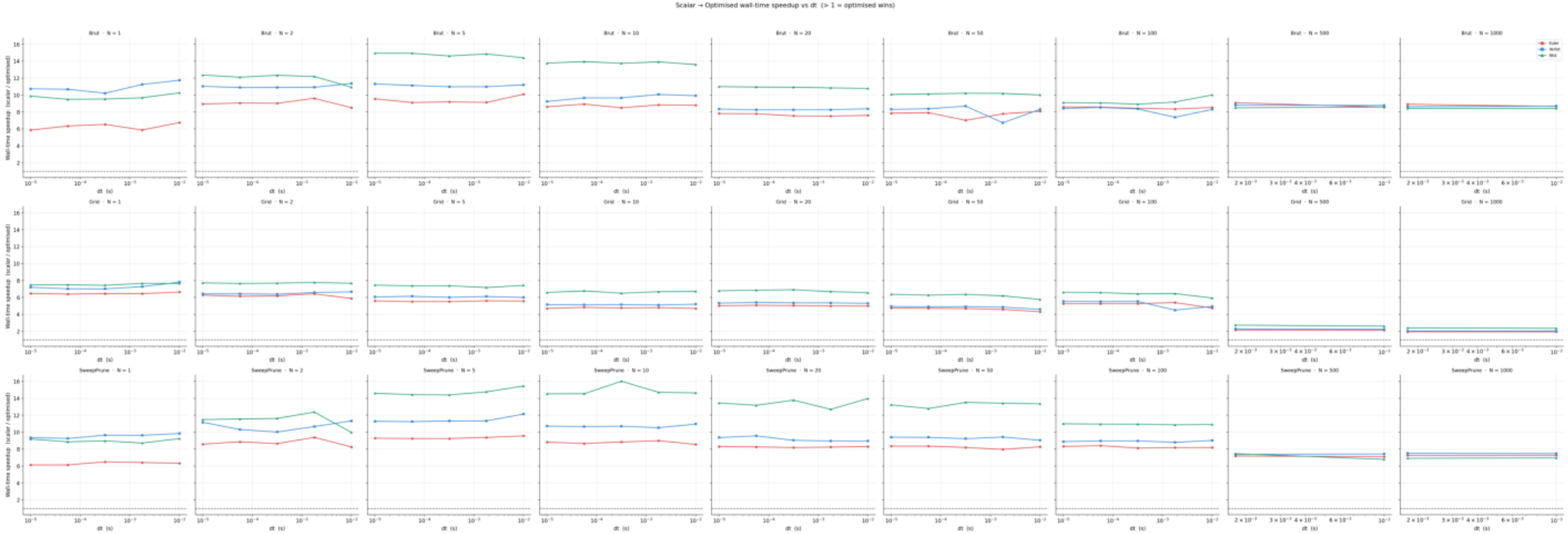

Wall-time scaling with object count
(dt = 1.78e-03 s)
Brut
Grid
SweepPrune
Wall time (s)
Number of objects
Number of objects
Number of objects
Euler
Verlet
RK4
scalar
optimised
0
5
10
15
20
25
0
200
400
600
800
1000

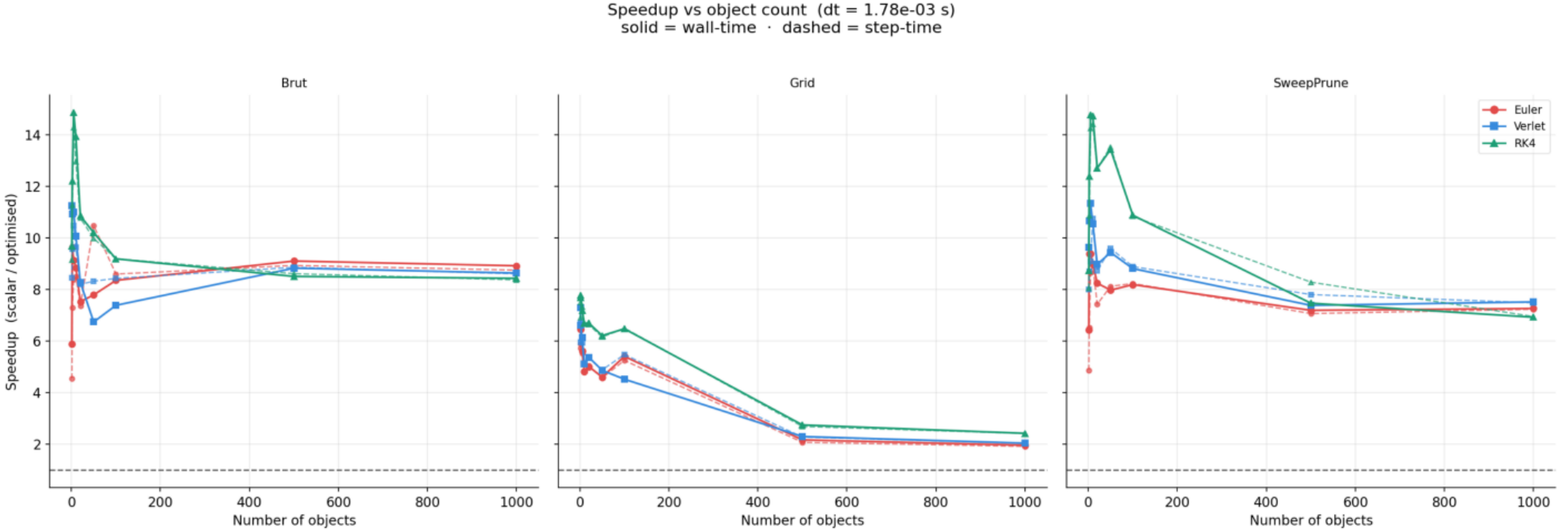

Step time vs dt
(plateau at fine dt = true per-step cost)
Brut · N = 1 objects
Brut · N = 2 objects
Brut · N = 5 objects
Brut · N = 10 objects
Brut · N = 20 objects
Brut · N = 50 objects
Brut · N = 100 objects
Brut · N = 500 objects
Brut · N = 1000 objects
Grid · N = 1 objects
Grid · N = 2 objects
Grid · N = 5 objects
Grid · N = 10 objects
Grid · N = 20 objects
Grid · N = 50 objects
Grid · N = 100 objects
Grid · N = 500 objects
Grid · N = 1000 objects
SweepPrune · N = 1 objects
SweepPrune · N = 2 objects
SweepPrune · N = 5 objects
SweepPrune · N = 10 objects
SweepPrune · N = 20 objects
SweepPrune · N = 50 objects
SweepPrune · N = 100 objects
SweepPrune · N = 500 objects
SweepPrune · N = 1000 objects
Step time (µs, profiling)
dt (s)
Euler
Verlet
RK4
scalar
optimized

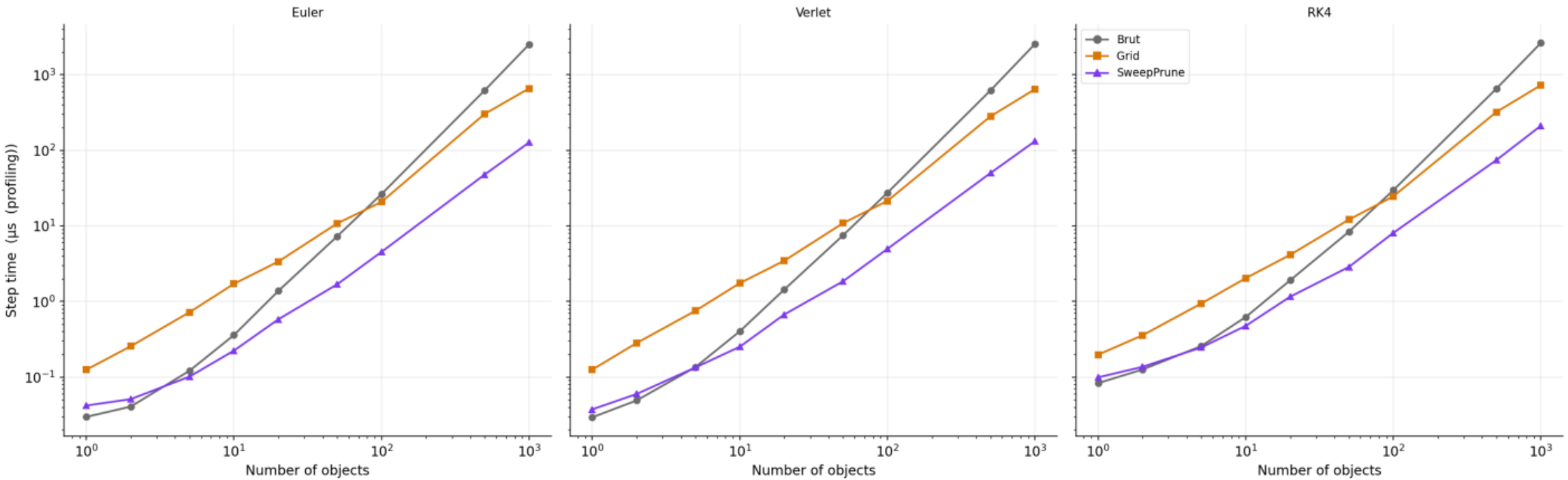

Step-time scaling with object count
(dt = 1.78e-03 s)
Brut
Grid
SweepPrune
Step time (µs (profiling))
Number of objects
Number of objects
Number of objects
Euler
Verlet
RK4
scalar
optimised
0
5000
10000
15000
20000
0
200
400
600
800
1000

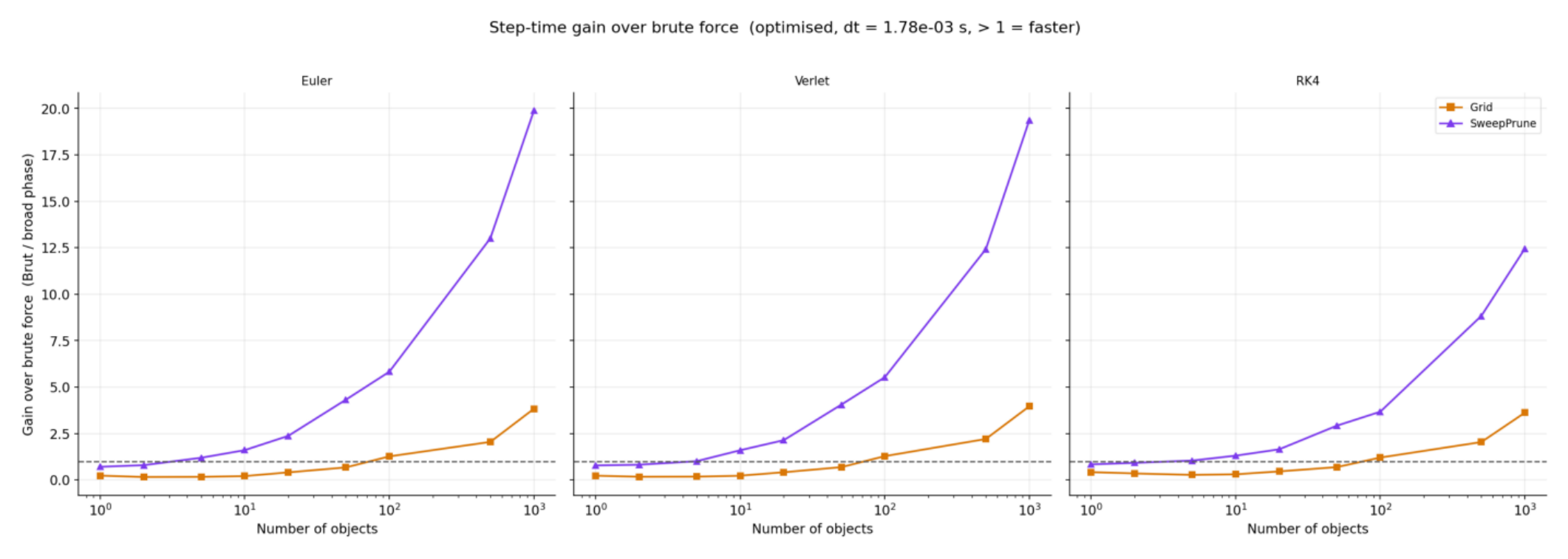

Per-step jitter vs dt (low CV → regular cost)
Brut · N = 1 objects
Brut · N = 2 objects
Brut · N = 5 objects
Brut · N = 10 objects
Brut · N = 20 objects
Brut · N = 50 objects
Brut · N = 100 objects
Brut · N = 500 objects
Brut · N = 1000 objects
Grid · N = 1 objects
Grid · N = 2 objects
Grid · N = 5 objects
Grid · N = 10 objects
Grid · N = 20 objects
Grid · N = 50 objects
Grid · N = 100 objects
Grid · N = 500 objects
Grid · N = 1000 objects
SweepPrune · N = 1 objects
SweepPrune · N = 2 objects
SweepPrune · N = 5 objects
SweepPrune · N = 10 objects
SweepPrune · N = 20 objects
SweepPrune · N = 50 objects
SweepPrune · N = 100 objects
SweepPrune · N = 500 objects
SweepPrune · N = 1000 objects
Step CV%
dt (s)
Euler
Verlet
RK4
scalar
optimized
5%
10% (high jitter)


Per-step jitter vs object count
(dt = 1.78e-03 s)
Brut
Grid
SweepPrune
Step CV%
Number of objects
Euler
Verlet
RK4
scalar
optimised
5%
10%


Scalar → Optimised wall-time speedup vs dt (> 1 = optimised wins)
Brut · N = 1
Brut · N = 2
Brut · N = 5
Brut · N = 10
Brut · N = 20
Brut · N = 50
Brut · N = 100
Brut · N = 500
Brut · N = 1000
Grid · N = 1
Grid · N = 2
Grid · N = 5
Grid · N = 10
Grid · N = 20
Grid · N = 50
Grid · N = 100
Grid · N = 500
Grid · N = 1000
SweepPrune · N = 1
SweepPrune · N = 2
SweepPrune · N = 5
SweepPrune · N = 10
SweepPrune · N = 20
SweepPrune · N = 50
SweepPrune · N = 100
SweepPrune · N = 500
SweepPrune · N = 1000
Wall-time speedup (scalar / optimised)
dt (s)
Euler
Verlet
RK4


Speedup vs object count (dt = 1.78e-03 s)
solid = wall-time · dashed = step-time
Brut
Grid
SweepPrune
Euler
Verlet
RK4
Speedup (scalar / optimised)
Number of objects
Number of objects
Number of objects


Step time by broad phase (optimised, dt = 1.78e-03 s)
Euler
Verlet
RK4
Brut
Grid
SweepPrune
Step time (µs (profiling))
Number of objects
Number of objects
Number of objects


Step-time gain over brute force (optimised, dt = 1.78e-03 s, > 1 = faster)
Euler
Verlet
RK4
Grid
SweepPrune
Gain over brute force (Brut / broad phase)
Number of objects
Number of objects
Number of objects
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
10^0
10^1
10^2
10^3